2. Project overview

2.1. Title of the project

The title of the project is Monkey Banana Adventure

2.2 Objective

The objective of the project is to create a simple ecosystem simulation using Scratch, including simple creative with game vibe. In this scenario, a monkey live in a jungle eats only banana and increase the score to the set points and goes to the next backdrop. The project want to show that in generally, The monkeys eat banana for living, for life in forest.

2.3 Target Audience

This project is designed for kids and beginner students who are learning about ecosystems and basic programming concepts in Scratch.

3.Features overview

3.1.Key features

_These are 5 characters such as Monkey , Banana, Jar, Ball and milk.

_The maximum number of bananas is:

#25 goes to the next backdrop2(name backdrop monkey2)

#50 goes to the next backdrop3(name backdrop monkey3)

#75 goes to the next backdrop4(name backdrop monkey4)

_When goes to next backdrop, the score set to zero.

_Monkey increases score every second when it eats banana.

_Monkey doesn't increase score when it eats jar, ball, and milk.

_Monkey goes randomly to eat banana.

3.2 Programming Concepts Used

_Variables (to eat banana, jar, ball , milk)

_If statements (to check if the monkey eats banana or not)

_Boolean operators(to check if the monkey is eating banana, jar, ball, or milk)

_Forever loop(to continuously such as jar or ball or milk)

_Collision detection(to detect when the monkey eating banana or other objects)

_Motion blocks(for random monkey movement)

_Random function(to move the monkey and objects to random positions)

_Backdrop switching(to change levels at 25,50,75 banana eating)

_Score reset mechanism(to set score to zero when backdrop changes)

_Sensing blocks(to detect eating conditions)

3.3. Ecosystems Network

This project represents a simple ecosystem interaction:

_Banana → Producer (provides food/energy)

_Monkey → Consumer (eats banana to gain energy)

_Jar, Ball, Milk →Non-living objects (do not provide energy and are not part of the food chain)

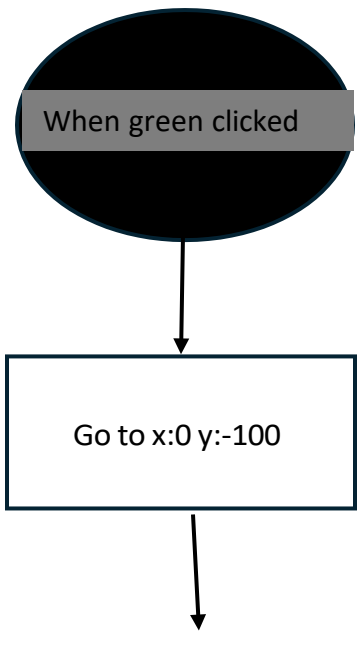
_If Monkey eats Banana →Monkey gains energy(score)

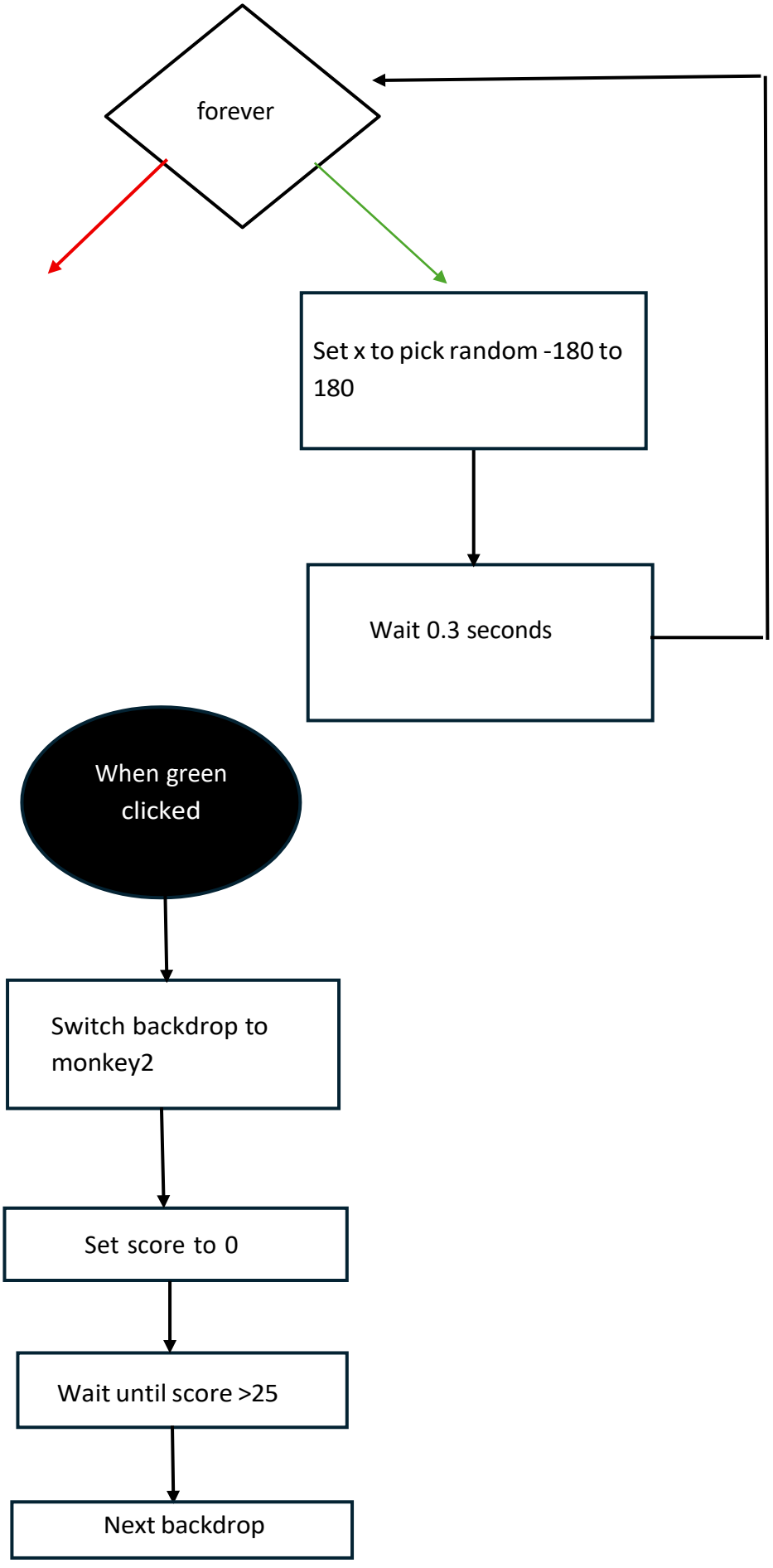
_If monkey eats Jar, Ball or Milk →Monkey does not gain energy(score)

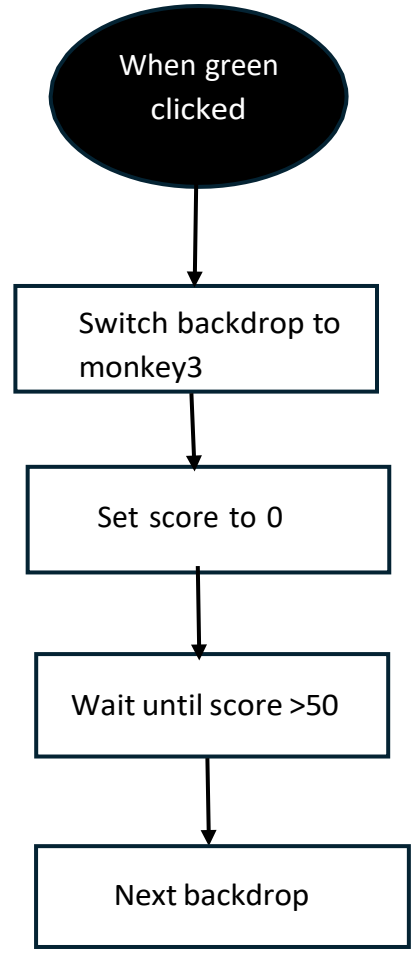
_Monkey live in jungle surroundings by natures

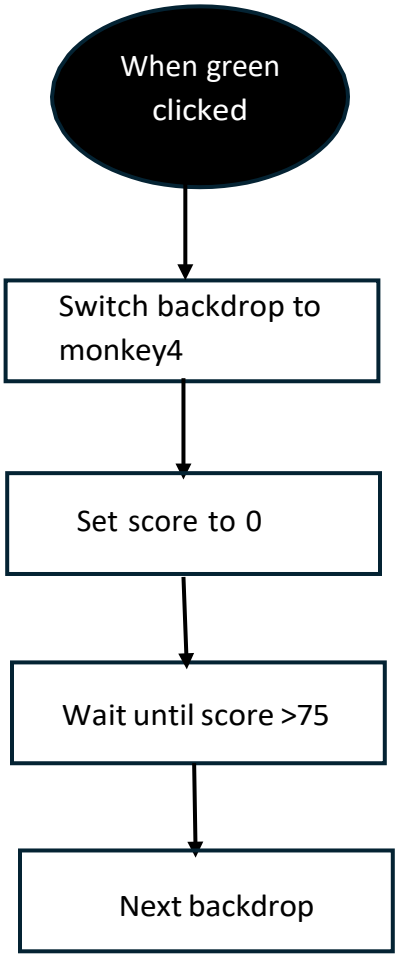
3.4. Algorithms & Flowcharts

A.monkey movement flowchart

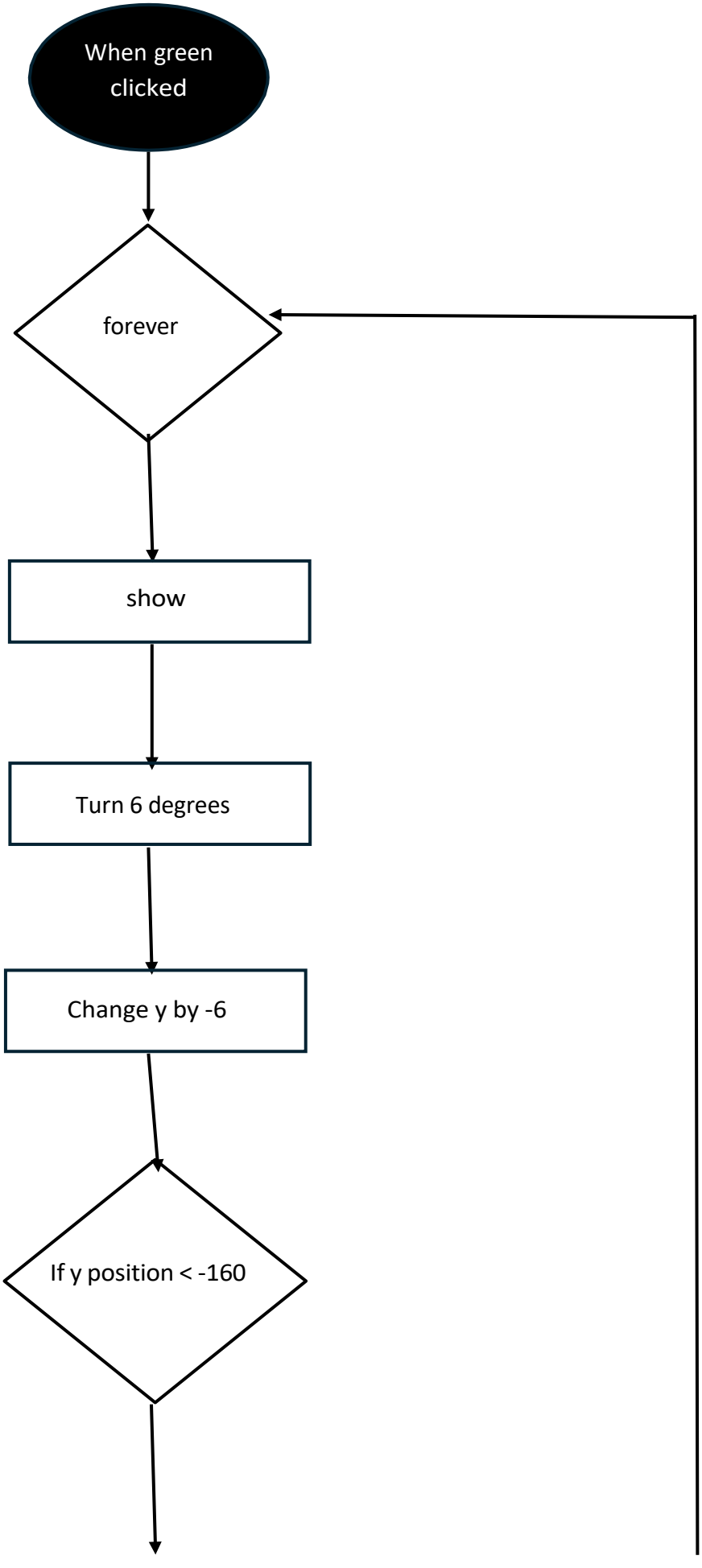


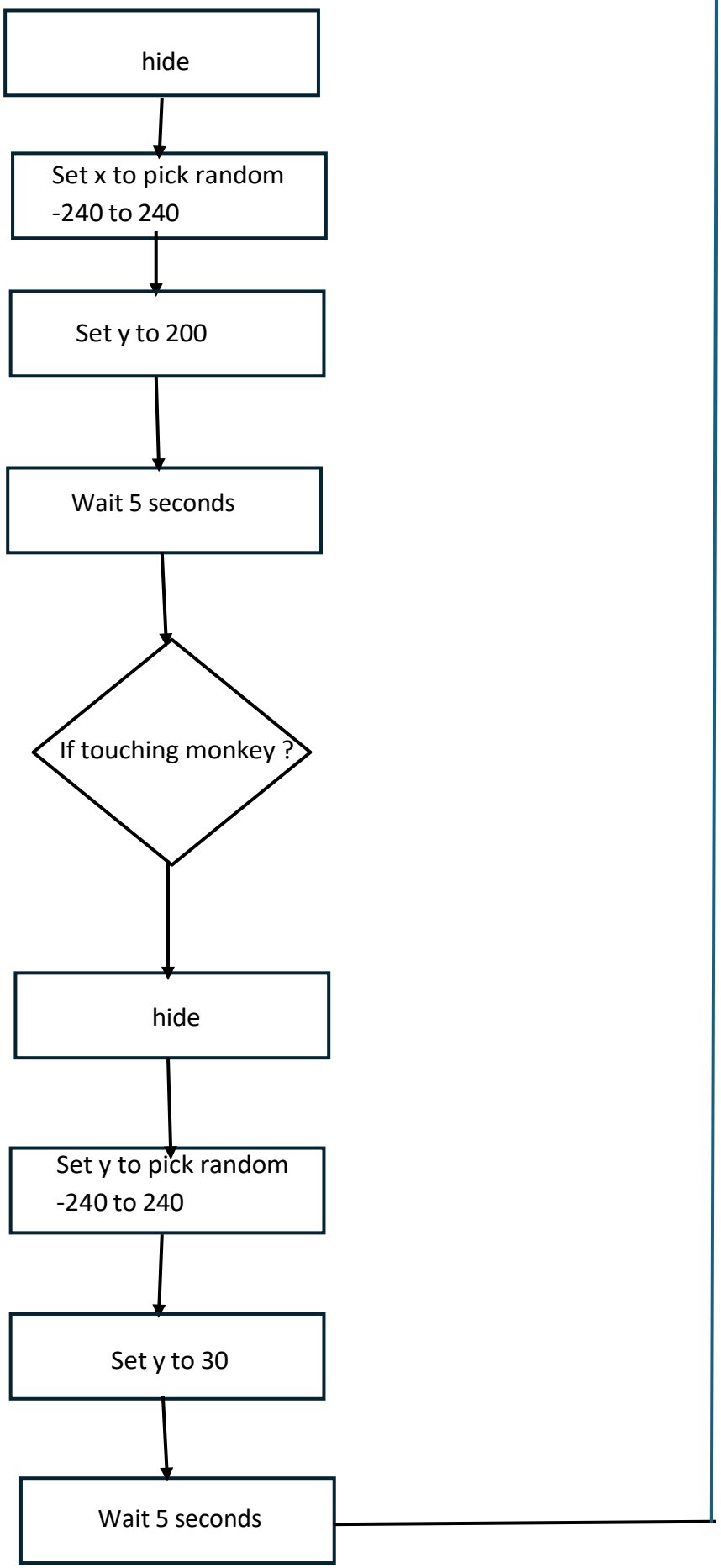

forever
Set x to pick random -180 to 180
Wait 0.3 seconds
When green clicked
Switch backdrop to monkey2
Set score to 0
Wait until score >25
Next backdrop


When green clicked
Switch backdrop to monkey3
Set score to 0
Wait until score >50
Next backdrop

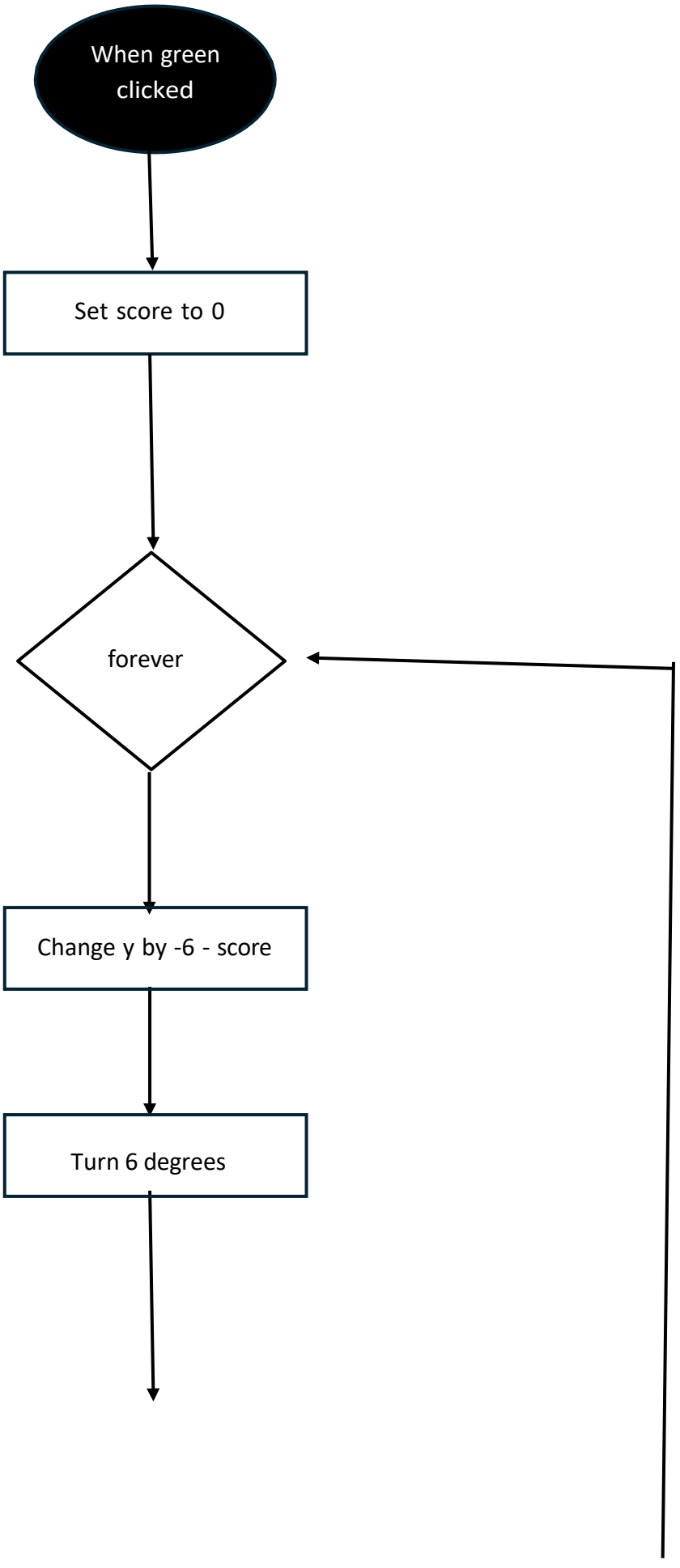

When green clicked
Switch backdrop to monkey4
Set score to 0
Wait until score >75
Next backdrop

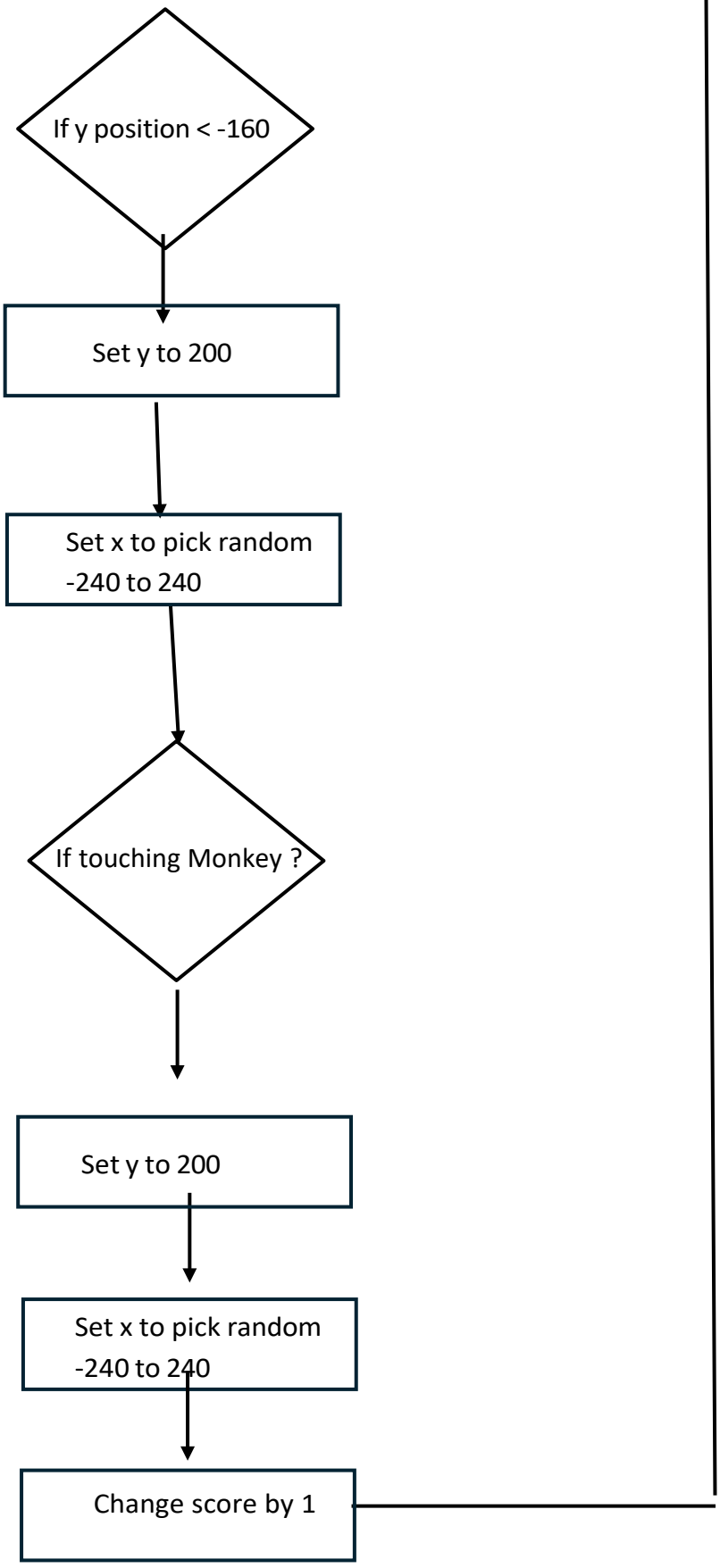
B.Jar flowchart



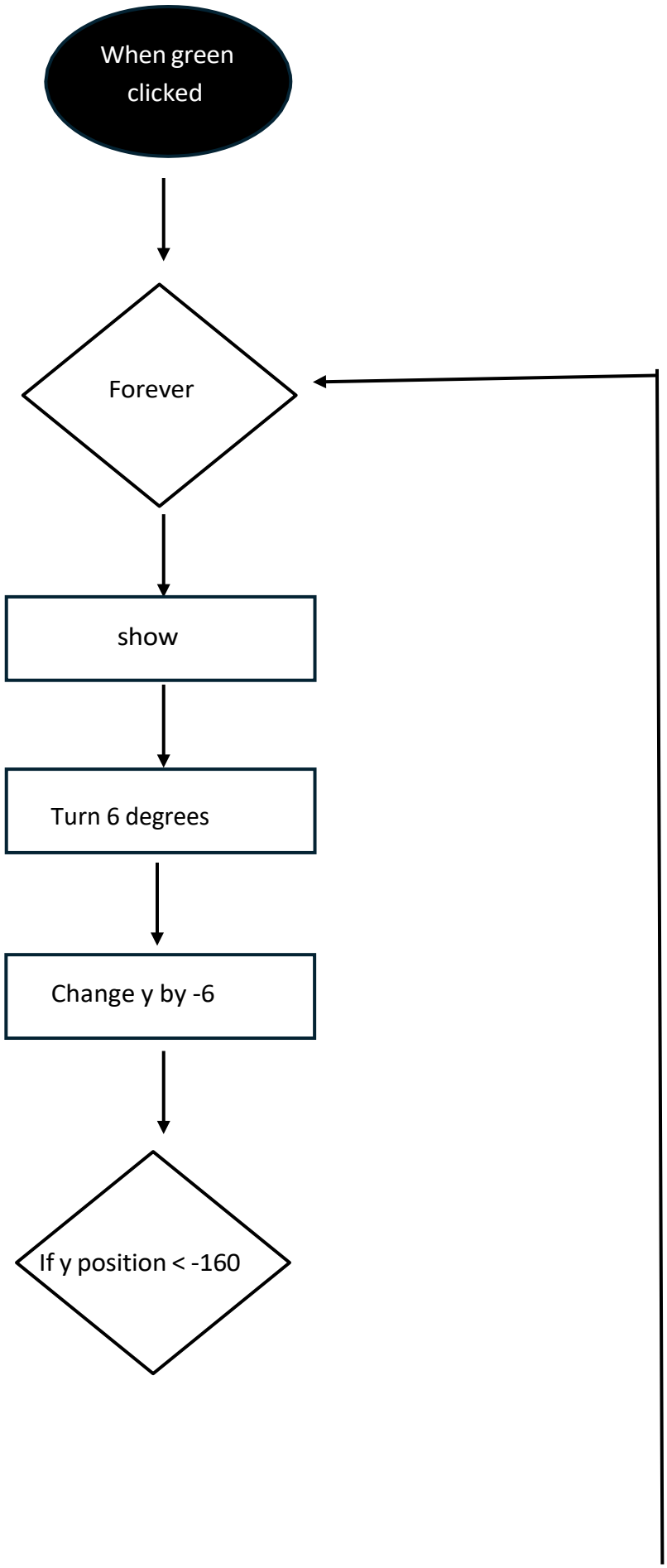

hide
Set x to pick random -240 to 240
Set y to 200
Wait 5 seconds
If touching monkey ?
hide
Set y to pick random -240 to 240
Set y to 30
Wait 5 seconds

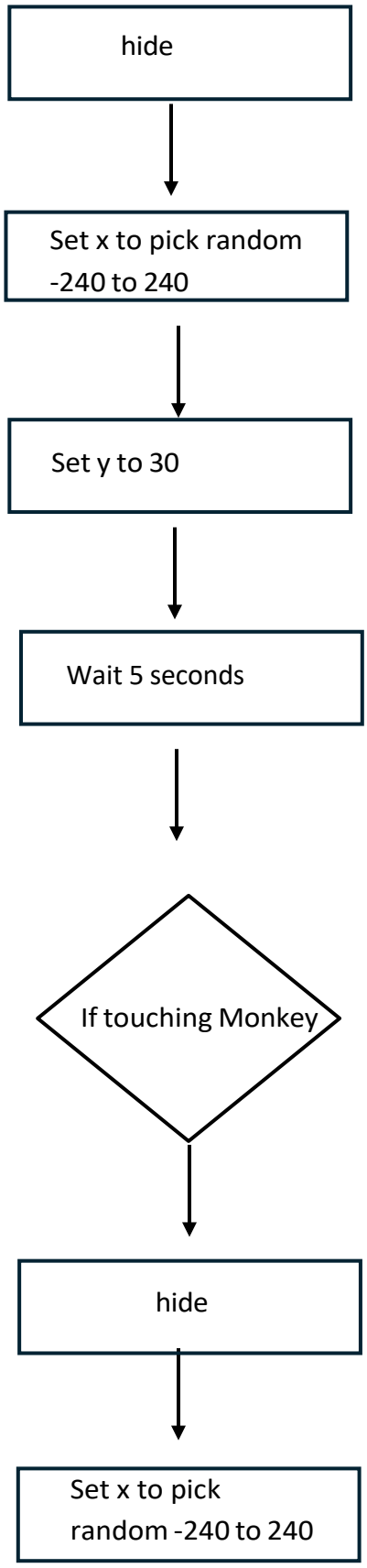
C. Banana flowchat

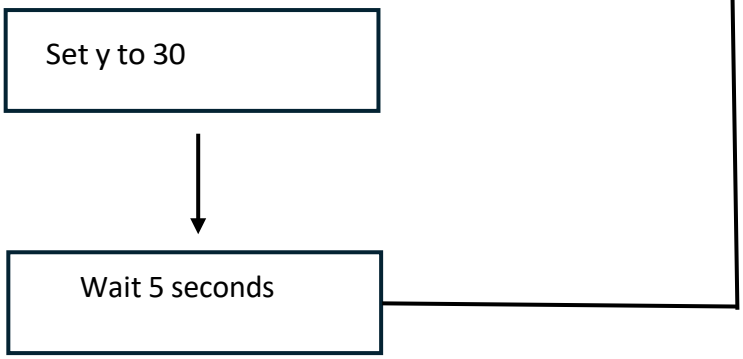



If y position < -160
Set y to 200
Set x to pick random -240 to 240
If touching Monkey ?
Set y to 200
Set x to pick random -240 to 240
Change score by 1

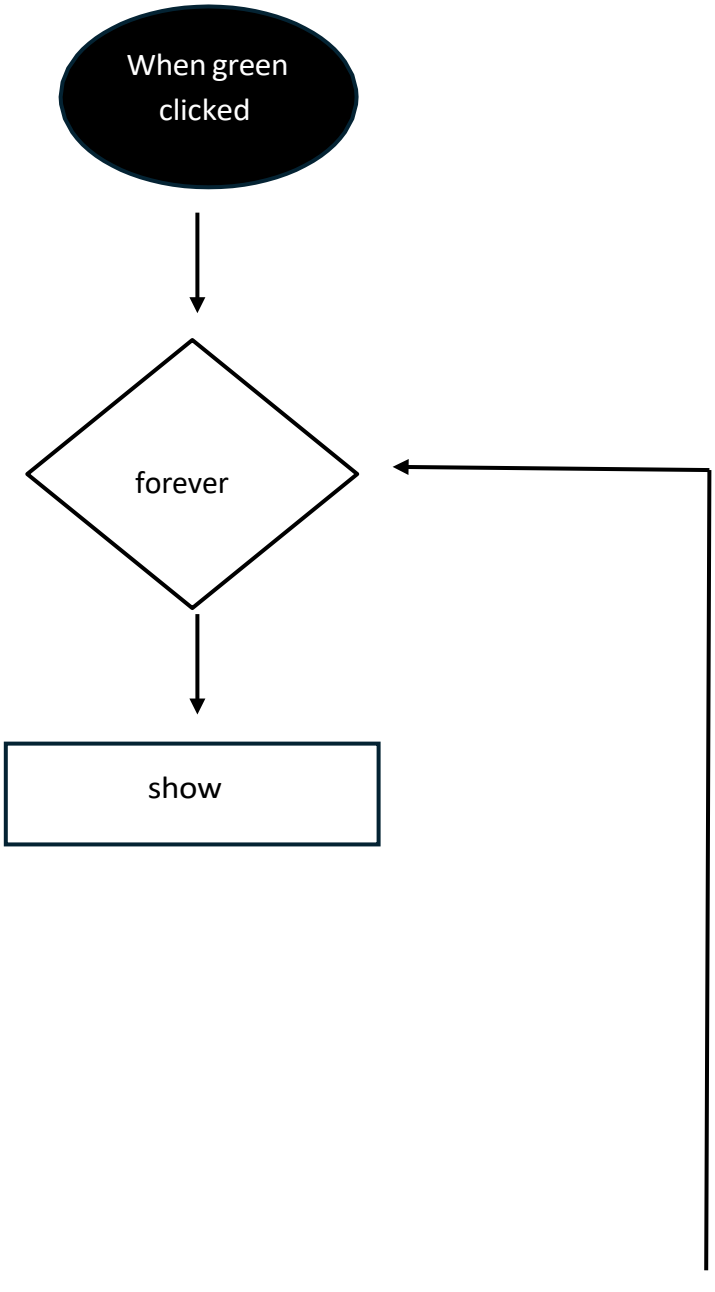
D.Ball flowchart

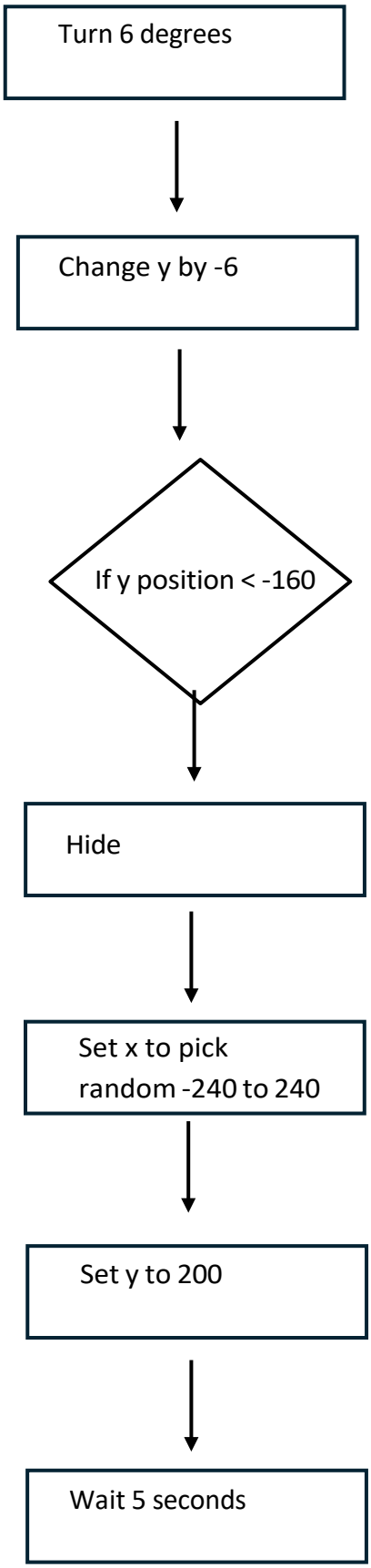


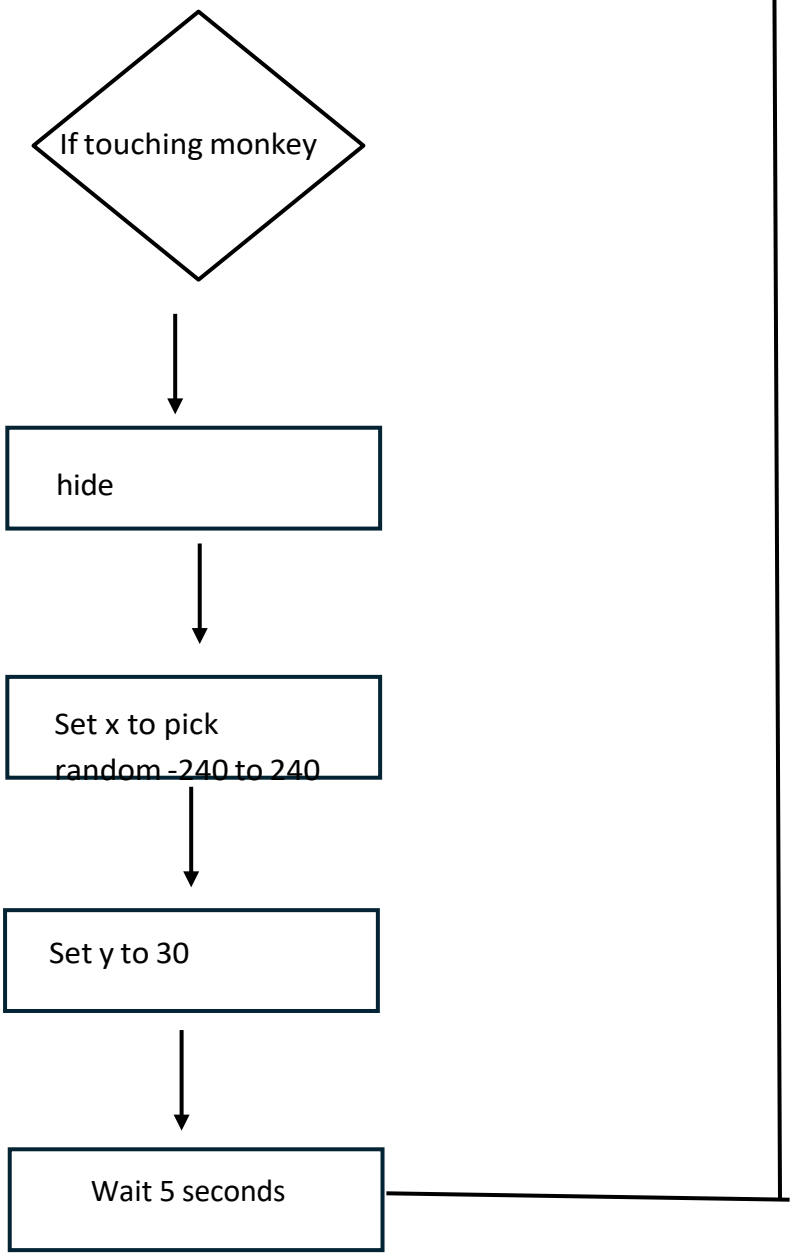

hide
Set x to pick random -240 to 240
Set y to 30
Wait 5 seconds
If touching Monkey
hide
Set x to pick random -240 to 240



E.Milk flowchart

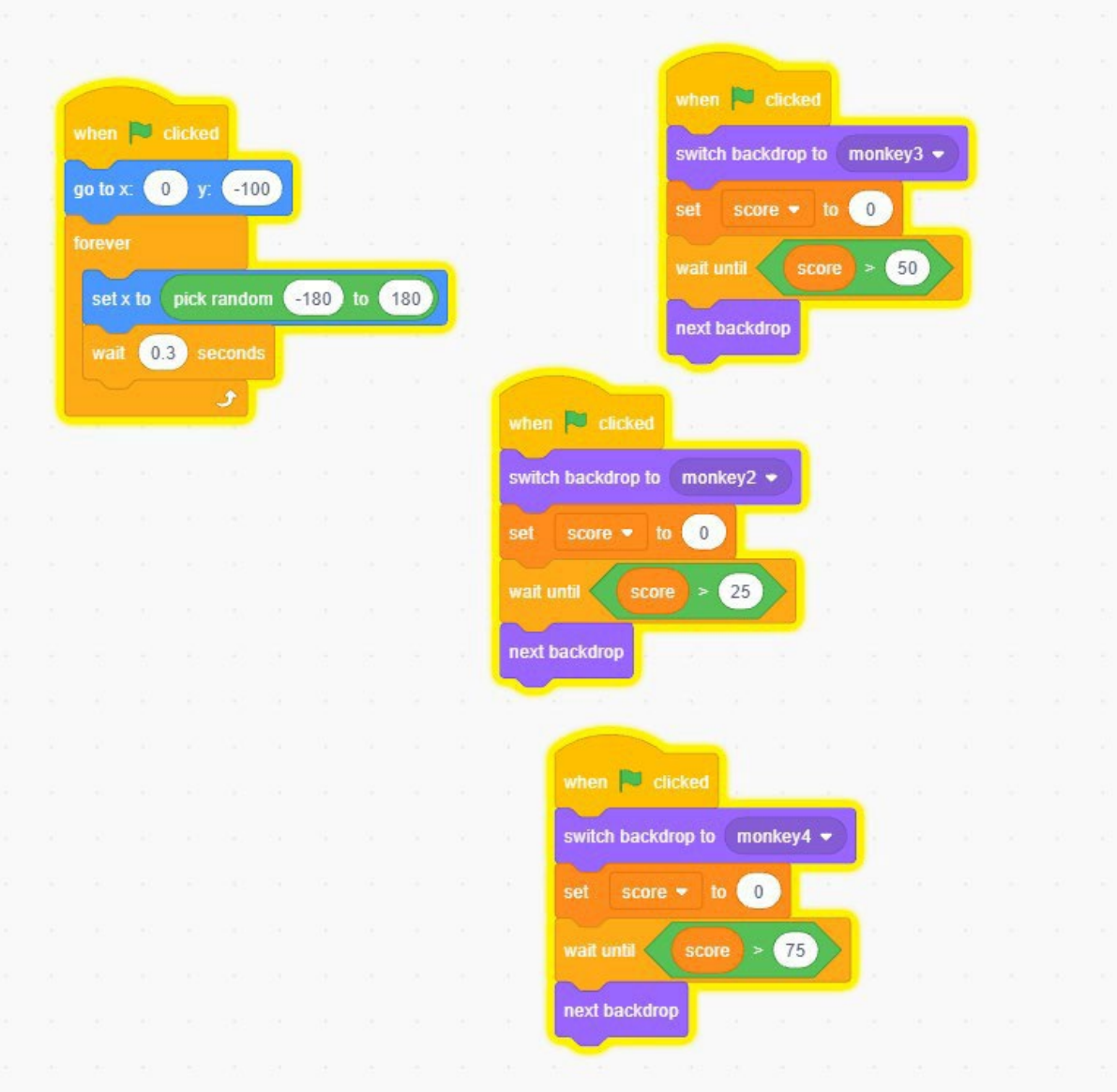



Turn 6 degrees
Change y by -6
If y position < -160
Hide
Set x to pick random -240 to 240
Set y to 200
Wait 5 seconds

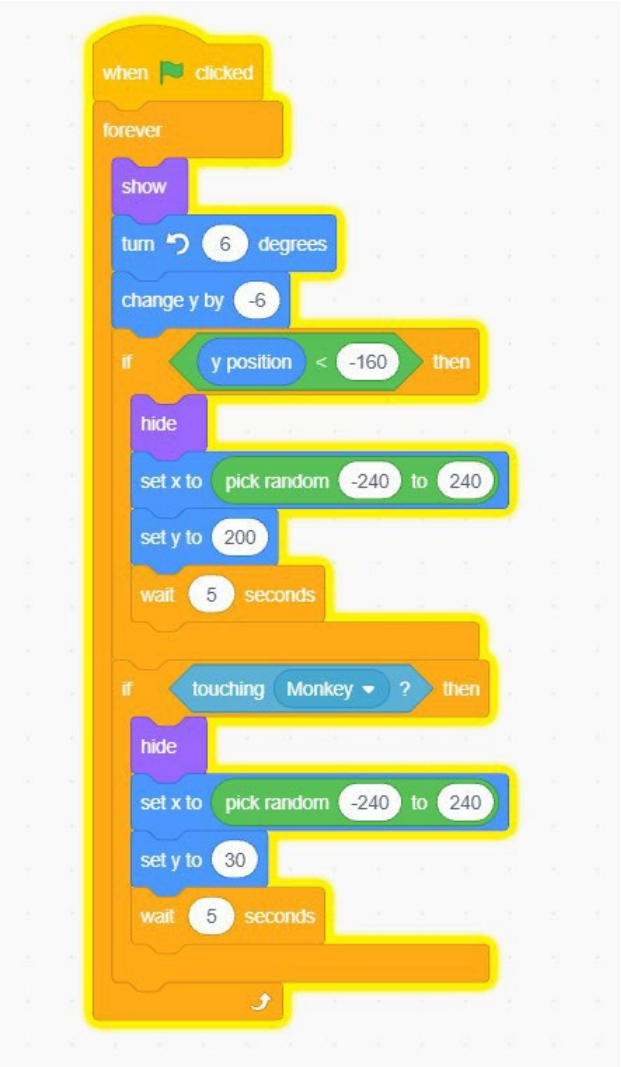

If touching monkey
hide
Set x to pick random -240 to 240
Set y to 30
Wait 5 seconds

#.Scratch code

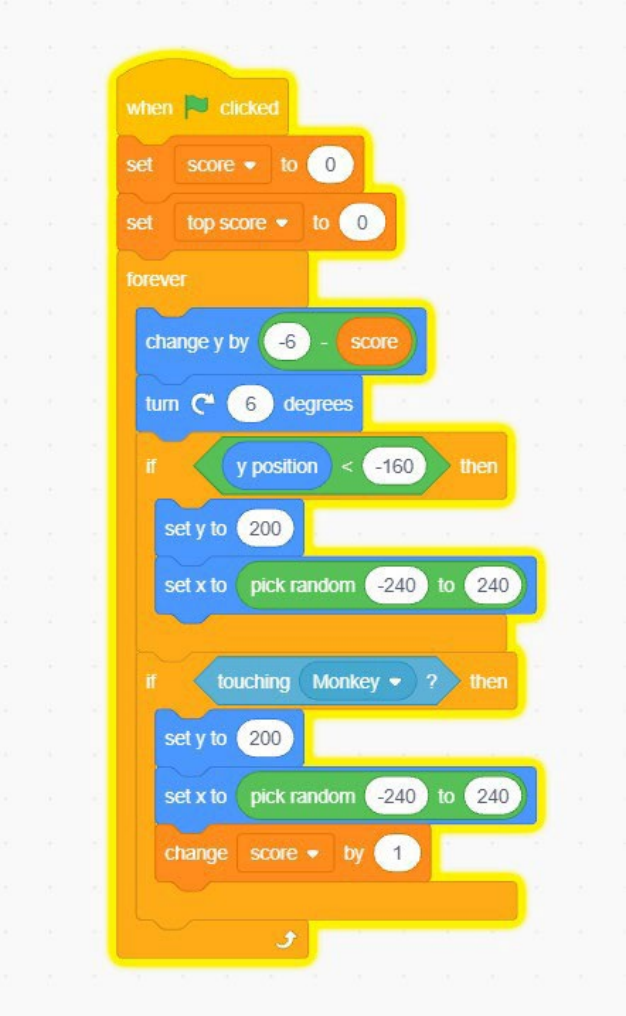
A. Monkey movement



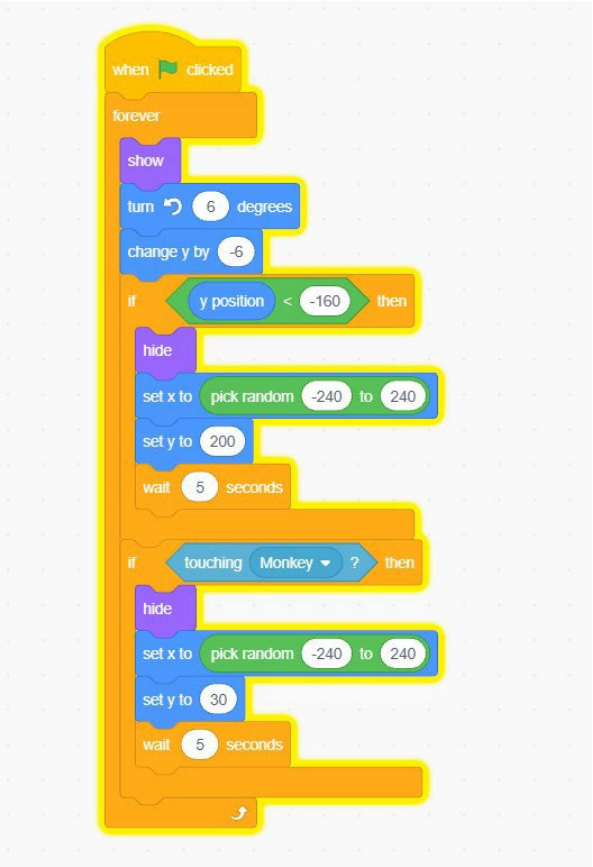
B. Jar code



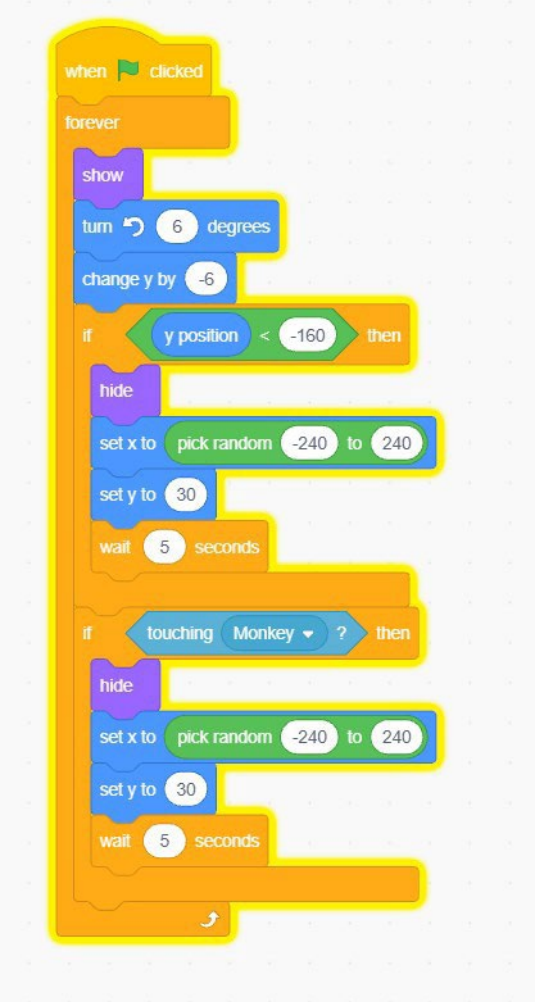
C. Banana code



D. Milk code



E. Ball code



3.5 learning Outcomes:

_What I learned:

- I learned how to create a game using Scratch
- I learned how to use variables(Score)
- I learned how to use if conditions to check touching objects
- I learned how to switch backdrops when a condition is true
- I improved my problem-solving skill by fixing error in movement and scoring

_Future improvements:

- Add sound effects when monkey eats banana
- Add a timer for the game
- Add more levels with different backgrounds
- Add a "Game ovet"screen

The end by Oeurn Meylih G5